R473: Extraction-joint redesign

TFPT RH programme

31 August 2026

Claim boundary. This note diagnoses one finite native read against the classical Guinand–Weil functional and redesigns the internal extraction joint. It does not prove compact-window positivity or the Riemann hypothesis. No statement about infinitely many k.

1 A — artefact diagnosis

The r470 sealed witness at $k = 5$ is the least eigen-direction of the native Hessian of `fullRead` on meshExp $= 3$, steps $= 24$ (`elementAnchor`$= 21 \le a_5 = 32$). Its native read is negative. The induced test function g_f is the even piecewise-linear autocorrelation of that direction (equal, for a piecewise-constant h, to the continuous autocorrelation).

The classical Guinand–Weil functional is computed at `mpmath` dps $= 50$:

- archimedean term A via the u-space digamma identity $A = C_b\, g(0) + \int_0^b w(u)\, [e^{-3u/2} g(0) - g(u)]\, du$, $w(u) = 2e^{-u/2}/(1 - e^{-2u})$, $C_b = -\gamma - \log \pi - \log(1 - e^{-2b})$;
- prime term P as the finite Λ-sum over $\log n \le \operatorname{supp} g_f = 3$;
- pole term Π the exact native-mesh pairing (`poleRead`, definitionally `weilPoleSide`).

The result is

$$Q(g_f) = A - P + \Pi = +0.0769078530458283 > 0,$$

while

$$\mathrm{fullRead}(a_5, m_5, f) = -0.0428854252772195 < 0.$$

The comb channel matches to machine precision ($P = 0.1364027472150502$). The entire gap is the arch tent error

$$\mathrm{err}_{\mathrm{arch}} = A_{\mathrm{classical}} - A_{\mathrm{window}} = 0.1197932783230478.$$

The cosine transform of g_f stays positive on a 120-point $[0, 30]$ grid ($\hat{g}_{\min} = 2.78 \cdot 10^{-7}$): g_f is a genuine autocorrelation. A Fejer-triangle ward gives $Q = +0.0880146104737050 > 0$.

Verdict A: ARTEFACT. Case (i) of the contract: $Q(g_f) \ge 0$ but fullRead < 0. The tent/window discretisation error at $k = 5$ exceeds Q. This is not a construction failure of g_f and not a Weil-negativity alarm.

The exact identity (Lean, proved, no digamma `sorry`) is

`fullRead\_weilForm\_gap\_eq\_arch`:
once `elementAnchor` $f \le a$, fullRead $-$ weilForm $=$ archRead $-$ weilArchSide.

Hence $|\mathrm{fullRead} - \mathrm{weilForm}| =$ `selectedArchError`. The chain needs this error term: on a PSD cap the redesigned readout is fullRead $\ge -\mathrm{err}_{\mathrm{arch}}$, not fullRead ≥ 0.

2 B — the provable polynomial bridge

The class A_{cap} can carry is the coefficient-plus-border space of real polynomials of degree < cap. Lean proves

> `selectedACapPsdImpliesPolynomialReads`:
> $A_{\mathrm{cap}} \succeq 0 \Rightarrow \mathrm{selectedReadQuadratic}(z) \ge 0$ for every $z \in \mathbb{R}^{\mathrm{cap}+1}$.

The channel is the moment evaluation (`PrimeWindow.hankel\_quadform`: $x^{\mathsf{T}} H x = \mu(p_x^2) - \nu(p_x^2)$; `PrimeWindow.A\_quadform` splits the border). There is no `steps` quantifier. Fixed k only.

The polynomial-to-`GridElement` gap is the named outer bridge

> `SelectedPolynomialApproximatesGrid`:
> at a fixed selected window, every mesh-compatible native read lies within $\mathrm{err}_{\mathrm{arch}}$ of the polynomial-class A_{cap} image.

On a PSD cap this is fullRead $\ge -\mathrm{err}_{\mathrm{arch}}$. It is a named Prop, not a `sorry`. At the sealed $k = 5$ witness $|\mathrm{fullRead}| = 0.042885425277 \le \mathrm{err}_{\mathrm{arch}}$.

3 C — redesigned joint

The historical FREQ endpoint `internal\_weil\_nonneg\_of\_frequently\_selected` is retained (it still consumes the r464 full-class channel as a hypothesis). The honest joint is now

> `FrequentlySelectedPolynomialJoint`
> = FREQ cone $\wedge$ `SelectedPolynomialApproximatesGrid`.

Fixed-k readout (proved as a function of the named approx):

> `weilForm\_ge\_neg\_two\_archError\_of\_joint`:
> at one good selected window, every onset-and-mesh-compatible f satisfies weilForm $f \ge -2\,\mathrm{err}_{\mathrm{arch}}(k, f)$.

No infinitely-many-k conclusion.

4 Joint map

Step	Status	Content
$A_{\mathrm{cap}} \succeq 0 \Rightarrow$ poly reads ≥ 0	proved	`selectedACapPsdImpliesPolynomialReads`
`fullRead-weilForm`= $\mathrm{err}_{\mathrm{arch}}$ (onset)	proved	`fullRead\_weilForm\_gap\_eq\_arch`
poly class $\to$ `GridElement` within $\mathrm{err}_{\mathrm{arch}}$	named	`SelectedPolynomialApproximatesGrid`
cone + approx $\Rightarrow$ fullRead $\ge -\mathrm{err}$	proved of named	`selected\_reads\_ge\_neg\_archError\_of\_approx`
onset $\Rightarrow$ weilForm $\ge -2\mathrm{err}$	proved of named	`weilForm\_ge\_neg\_two\_archError\_of\_joint`
historical FREQ $\Rightarrow$ `weilForm`≥ 0	retained, conditional	still needs the obstructed full-class channel

5 Kill table

Candidate	Status	Reason
Negative $Q(g_f)$ / ALARM	killed	$Q = +0.0769078530458283$; $\hat{g} \geq 0$.
g_f not an autocorrelation	killed	PWC ACF = PL interpolant; $\hat{g}_{\min} > 0$.
Restore the r464 full-class channel	killed	r470 obstruction stands; residual $24 > 11$.
Cone $\Rightarrow$ native fullRead ≥ 0	killed	artefact: tent error $> Q$ at $k = 5$.
New infinitely-many-k read-out	killed	r469 anti-list; joint is fixed-k.
R473 proves RH	killed	finite diagnosis + poly PSD; census stays 8.

6 Census

Lean `sorry` census remains 8: three in `Canonical.lean`, one named Gauss–Mellin–digamma identity in `Elementwise.lean`, three external arrows in `ExternalBridges.lean`, and one source theorem in `Source.lean`. No new `sorry`. Smoke 13/13, `SPEC_SHA 0650125bd812c1f8`.